

Antiproliferative effects of apple peel extract against cancer cells.

Studies have shown an inverse relationship between the consumption of apples and the risk of several cancers. The peels of apple, which have been shown to possess exceptionally high concentrations of antioxidants, are often discarded. In this study, we evaluated the antiproliferative effects of apple peel extract (APE) in variety of cancer cell types. Our data demonstrated that APE, obtained from organic Gala apples, imparted significant reduction in the viability of a variety of cancer cell lines. Further, our data showed a significant decrease in growth and clonogenic survival of human prostate carcinoma CWR22Rnu1 and DU145 cells and breast carcinoma Mcf-7 and Mcf-7:Her18 cells. Also, the antiproliferative effects of APE were found to be accompanied by a G0-G1 phase arrest of prostate and breast cancer cells. Furthermore, APE treatment resulted in a marked concentration-dependent decrease in the protein levels of proliferative cell nuclear antigen, a marker for proliferation. In addition, APE treatment resulted in a marked increase in maspin, a tumor suppressor protein that negatively regulates cell invasion, metastasis, and angiogenesis. Our data suggested that APE possesses strong antiproliferative effects against cancer cells, and apple peels should not be discarded from the diet. Detailed mechanistic studies, especially in appropriate in vivo animal models, are needed to further examine the antiproliferative and preventive effects of APE against cancer.